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RESEARCH ARTICLE

Hierarchical Multi-Class and Multi-Label Text Classification for Crime Report: A Traditional Machine Learning Approach

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ABSTRACT Large amounts of digital data are produced daily through society's use of government and private companies. Digital transformation contributes to the increasing amount of structured and unstructured data stored in digital media. Organizations build centralized data repositories to store and provide information to business areas, supporting Business Intelligence solutions. Some databases store vast amounts of unstructured data, which must be systematized and classified to meet the data owner's needs. In criminal incident report systems, each recorded incident must be classified as a specific crime, with hundreds or thousands of categories presented to the responsible officer. This work explores a clustering approach to group categories into a hierarchical tree of classes, enabling the use of Machine Learning (ML) models like XGBoost for automated classification of criminal incident reports narratives. As a case study, the Civil Police of the State of Rio de Janeiro (SEPOL/RJ) has a database with over 6.5 million records, growing daily from Judicial Police Units (JPU) across the state. Each new report requires manual classification. A hierarchical tree of classes was developed to segment the problem, allowing various XGBoost models for automated classification. The proposed hierarchical model with 80 classes achieved an accuracy of 0.463, outperforming the baseline flat model which reached 0.419, along with a 25.48% reduction in training time. The weighted average F1-score obtained by the hierarchical model was 0.48188, while the baseline model reached 0.44061. The improvement was statistically validated through a Wilcoxon signed-rank test, which yielded a p-value of 0.000010.

INDEX TERMS Hierarchical multilabel text classification, crime narratives, XGBoost, text embeddings, semantic clustering, public security data.

I. INTRODUCTION

Law enforcement agencies worldwide record a large volume of criminal incident reports daily. These records are essential for crime analysis and prevention. However, many of these data are unstructured, challenging direct analysis and efficient use. For these data to be properly utilized, they must be classified into specific types of crime according to the

criminal code of each country. Moreover, this classification is crucial not only for using the data in analyses but also for ensuring the Due Process of Law for each incident. Proper categorization of crimes is vital to ensure that the legal process follows established procedures, allowing investigations and legal actions to be carried out accurately and fairly ([1], [2]).

The categorization of criminal incident narratives is essential for the effective management of public security, as it transforms large volumes of unstructured data from

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daily police reports into strategic information. By organizing these narratives, it becomes possible to identify patterns and map areas with higher crime rates, facilitating the planning of targeted and specific actions by law enforcement agencies. This analysis not only enables the development of more effective crime prevention and control policies but also promotes the integration of information across different government sectors. Basílio et al. ([3] and [4]) propose a LDA-based text mining method for the classification of public security data from Rio de Janeiro, Brazil, allowing for evidence-based decision-making and more accurate policing strategies tailored to the reality of each community.

While classifying the narratives of these incident reports, a police officer often faces the challenge of choosing from thousands of possible classes, making the task difficult. This wide range of options can make the process time-consuming and tiring, increasing the likelihood of human error. Furthermore, selecting the correct class requires a detailed understanding of the characteristics of each type of crime, which is not always easy to identify in the narratives, especially when the descriptions are vague or ambiguously written. This scenario can undermine the accuracy of classifications and, consequently, impact the efficiency and reliability of subsequent analyses ([5], [6]).

Recent efforts in hierarchical text classification have explored a range of modeling strategies, from traditional machine learning to deep learning. Kusakin et al. ([7]) investigated hierarchical classification using BERT-based models applied to structured scientific texts in Russian. Their work focused on abstracts, titles, and keywords, which follow standardized formats and domain-specific terminology. More recently, a modular framework for Hierarchical Article Classification (HAC) was proposed by Garcia et al. ([8]), comparing classical algorithms such as SVM with transformer-based models like BERT, Llama3.1, and GPT-4. Applied to scientific publications from Fiocruz, this framework demonstrated that traditional models can match or even surpass deep learning models in certain domains, particularly in the classification of texts within the Biological and Health Sciences. These findings support the notion that lightweight and interpretable models remain relevant in hierarchical text classification.

In contrast to these works, which rely on structured and domain-specific text, our study addresses the challenges of classifying unstructured, emotionally charged criminal narratives. These texts are often written under stress, using informal language and lacking standardization. This condition influences the narrative description, making it less objective or clear. Similarly, the officer responsible for the classification may be fatigued, especially when working a night shift or after an intense day filling numerous incident reports. This physical and mental exhaustion can affect their ability to analyze and judge, making the classification process even more prone to errors. These human variables – both in the citizen's report and in the officer's work – create gaps

that can undermine the accuracy of the classification system, leading to inconsistencies and flaws in the process. In this case, an automated system for classifying the narratives can be a significant factor in the success of both data and judicial analyses ([9], [10]).

The literature addresses three main concerns in AI-based text classification of criminal incident reports: accurate crime narrative classification, AI applications in law enforcement, and organizing information for text classification. The first concern is to correctly classify a crime narrative into the appropriate type of crime, as seen in Sarzaeim et al. [11] that focus on classifying crime reports in high-crime cities like San Francisco and Los Angeles, using AI and Deep Learning techniques to improve public service. Another critical concern revolves around the potential biases in AI models used for law enforcement, where AI can help mitigate human biases in sentencing, especially regarding gender and race. Ho et al. [12] examine AI's role in reducing bias in judicial sentences for repeat offenders, showing that AI can outperform humans by eliminating biases during data preprocessing. The third concern is finding ways to organize the data in a manner that enables and improves crime narrative classification algorithms, Ku et al. [13] present a three-step approach for organizing large crime report datasets, using text preprocessing, entity analysis, and Naive Bayes classification for efficient crime report classification.

The literature also brings various approaches to text classification, including word frequency methods for theme extraction, as seen in Das and Das [5], who combine this with graph construction for hierarchical classification. Another approach involves Named Entity Recognition (NER) and graph construction, where texts are classified based on clusters formed through centralities, as shown in Das and Das [10]. Additionally, text classification is applied in crime detection in CCTV footage, where frames are processed by a Large Language Model (LLM) to generate image-to-text descriptions, which are then analyzed to identify the type of crime in the images [14].

Some countries have Crime Reporting Systems that function as emergency hotlines to receive reports and relevant crime information. Typically, these lines are called tip lines, anonymous tip systems, or hotlines. In the state of Rio de Janeiro, Brazil, there is a hotline called “Disque Denúncia” (DD). Some published works in the literature have used Machine Learning (ML) techniques to classify parts of this hotline data. The work in [9] presents unweighted neural networks for classifying text from complaints in this hotline subsystem. This subsystem receives complaints handled by humans, recorded, and classified into primary and secondary categories.

Unlike previous studies, this work adopts a hierarchical classification approach, in which each level of the decision tree requires specific models to make predictions at different stages. Several XGBoost models were trained and optimized to maximize both accuracy and F1-score for each hierarchical

decision. Through this process, the best models for each level and decision were selected, considering the relevance and context of each choice. The performance of the proposed approach achieved an accuracy of 0.463, compared to 0.419 from the naive approach, and a weighted average F1-score of 0.48188, compared to 0.44061 for the flat baseline model, while also reducing the training time by 25.48%. These results demonstrate the feasibility and effectiveness of hierarchical multi-label classification in the context of SEPOL/RJ data.

The original dataset contains 1,170 distinct crime types recorded by SEPOL/RJ. However, we reduced this number to 80 representative classes that together cover approximately 90% of all cases. This selection was based on the frequency distribution of the crimes, allowing us to focus on the most prevalent types while maintaining a manageable number of classes for modeling. Classifying all 1,170 crime types in a single model would be computationally unfeasible and conceptually inadequate, given the multifaceted nature of the data and the high class imbalance. By concentrating on the 80 most frequent classes, we ensure broad coverage of the dataset while enabling the construction of a robust and interpretable hierarchical classification system.

This decision to use the XGBoost model as an initial approach for classifying crime report narratives was based on its simplicity, lower infrastructure requirements, faster training time, versatility, and efficient processing. It provided a practical and cost-effective way to validate our proposal and build a solid foundation for future work. Additionally, the sensitive nature of the data used in this study imposes strict privacy constraints, preventing external processing or cloud-based solutions. Many institutions also lack access to GPU infrastructure, making deep learning approaches less feasible in practice. While transformer-based models such as BERT and GPT have shown strong performance in processing unstructured text, recent studies demonstrate that classical machine learning algorithms can also be highly effective. Traditional machine learning algorithms such as SVM, XGBoost, decision trees have achieved comparable or even superior results in certain scenarios ([8]). These models offer better interpretability, lower computational cost, and greater adaptability to resource-constrained environments, aligning well with the practical needs of real-world deployments.

In a nutshell, this work makes five main contributions to the field of criminal incident report classification:

- It presents an innovative methodology for categorizing criminal incident narratives, facilitating their organization and analysis.
- It proposes a machine learning-based approach that demonstrated an accuracy improvement of 5%, a weighted average F1-score improvement of approximately 9.4% (from 0.44061 to 0.48188), and a training time reduction of 25.48% compared to naive flat classification methods.
- It validates the performance improvements through statistical analysis using the Wilcoxon signed-rank test,

which confirmed the significance of the F1-score gains per class (p -value = 0.000010).

- It brings an effective strategy for extracting valuable insights from large volumes of unstructured data, even with a high number of possible classes.
- It develops a practical application for a case study involving the Civil Police of the State of Rio de Janeiro (SEPOL/RJ), enabling a prediction probability-based model to assist in improving the classification of crime reports.

This article is structured as follows. After this introduction addressing the challenges and importance of classifying criminal incident reports, Section II explores related works, providing an overview of existing approaches for classifying and grouping criminal data. Section III describes the materials and methods developed and employed in this study, detailing the procedures and techniques used. Section IV presents the experiments, the metrics and parameters applied, the analysis of results, and a discussion of their implications. Finally, Section V contains the conclusion of the work, along with suggestions for future studies and improvements in the field.

II. THEORETICAL BACKGROUND

A. TEXT CLASSIFICATION

Text classification is the process of grouping texts with similar characteristics under the same label, known as a class. The text classification problem can be binary or multi-class, indicating that there may be only two or several possible classes to be assigned to the text body. It can also be single-label or multi-label. Multi-label problems allow the model to classify the given text into more than one class [15].

Sakhrani et al. [16] present a text classification approach by transforming a potential multi-label problem into a single-label one. The authors use a technique to pre-group labels that frequently appear together into a new single label. The study also introduces using LLM to augment the queried text by adding extra information. The process is similar to a researcher reading an article's abstract but with an additional background of tacit knowledge from the literature. Subsequently, the queried text and the extra information are classified into one of the available labels.

Multi-label text classification involves assigning more than one class to a text based on analyzing the terms in the text body. Li et al. [17] presents an open-world multi-label text classification. In this type of task, the possible classes are not available *a priori*. The method needs to perform topic modeling for subsequent text classification. So, it fragments the texts and sends them to an LLM for keyword extraction. The words are grouped, and the centroids of each group form the labels for text classification.

The labels for the text classification procedure can all be available *a priori*. However, some researchers investigate the need for a dynamic update of the possible labels available for classification. Das et al. [18] work with a closed set of labels. However, as new entries are presented for classification,

the employed method checks whether the new text fits into one of the available classes. If it does not, a new class is created, and the text is classified. From that point on, the new label becomes available for classifying other texts that are subsequently submitted.

The literature presents works that use text classification to detect whether a text was generated by AI, in which case a binary classification is performed. In [19], the authors use LLMs to perform the classification along with the probability that the procedure was carried out correctly. From the results of several models, an input is constructed for the use of traditional ML, characterizing the model ensemble process.

Shen et al. [20] present a method for structuring multiple possible classes for text classification in a hierarchical tree. Using this tree, the authors perform the classification by following a path through the nodes to the leaves. The response classes consist of those present in the path taken for classification. In this case, a way of constructing a hierarchy from the available classes into a tree is presented.

Bashatah and Sherry [21] develop a methodology for classifying Standard Operating Procedure (SOP) manuals using Prompt Engineering and chat LLMs. The work aims to categorize them into groups to reduce the time needed to search for information in emergencies. The authors also introduce the Available Operational Time Window (AOTW) concept to determine the time for consulting the respective manuals.

Kusakin et al. ([7]) investigated BERT-based models trained on structured scientific texts in Russian and identified several factors that impact performance. Their findings show that domain-specific pretraining significantly improves classification quality compared to general-purpose language models. They also observed gains when fine-tuning deep layers of the transformer and when concatenating titles with annotations. However, architectural changes such as increasing linear layer dimensions or adding skip-connections had minimal effect. Additionally, they demonstrated that class imbalance impacts F1 scores negatively for smaller classes.

In contrast, Garcia et al. ([8]) evaluated a modular hierarchical classification framework applied to scientific publications from Fiocruz, testing various combinations of feature extraction methods, hierarchical classification strategies, and algorithms. Configurations using traditional models, particularly SVM combined with TF-IDF and the Local Classifier per Parent Node (LCPN) strategy, achieved the best overall results, outperforming some transformer-based alternatives such as BERT, GPT-4, and Llama3.1. Together, these studies reinforce the idea that classical machine learning models remain competitive in hierarchical classification tasks and can, in many scenarios, match or exceed the performance of more complex deep learning approaches.

B. WORD EMBEDDINGS

Techniques for text representation in numeric vectors have evolved over time, keeping pace with the increasing volume and complexity of textual data [22]. Initially, approaches were word counting, based on counting the frequency of words relative to the overall context of the text. Included in this early era of text vectorization are techniques such as one-hot encoding [23], TF-IDF (term frequency-inverse document frequency) [24], and bag-of-words [25]. These techniques were used to transform text into numerical data to be processed in ML models [22].

The early approaches presented in the literature generated vectors with high sparsity, which made it difficult to train the models later. Another important point was the lack of consideration of the semantic context since these techniques were word-counting-based, focusing solely on word relative frequencies. Techniques such as Latent Semantic Analysis (LSA) [26] and Latent Dirichlet Allocation (LDA) [27] attempted to address sparsity using statistical methods. However, they were still limited in capturing deep semantic relationships [22].

With the advancement of ML, prediction-based models emerged, which learn text representations through models such as Word2Vec [28]. That model was developed by Google, marking a milestone in understanding the semantic meaning of words. Word2Vec generates dense vectors that capture semantic relationships between words using models like Continuous Bag of Words (CBOW) and skip-gram. CBOW model predicts the central word given the context, while the skip-gram model predicts the context given a central word [22].

The next step in the evolution of text vectorization occurred with the emergence of the Transformer architecture [29] and the use of Transfer Learning. Transformers allow for modeling the context around a word more properly, generating vector representations that capture the word's semantic meaning in the context used [22]. Transformers are divided into two types. The first is Bidirectional Encoder Representations from Transformers (BERT), which uses only the encoder part of the Transformer architecture. The second class is the Generative Pre-trained Transformer (GPT), which uses only the decoder part of the structure mentioned above.

The “mE5-large-instruct” is a multilingual transformer-based model provided by Microsoft [30]. The model performed better compared to other models trained only in English. This model is a variation of the “mE5-large” adjusted with additional synthetic data generated by GPT-3.5/4. A training approach based on instructions was used in “mE5-large-instruct”, allowing the model to better understand the proposed task and generate better text vectors. The chosen model demonstrates excellent performance in multilingual benchmarks, including MIRACL and bitext mining, making it the most suitable.

C. THE XGBoost MODEL

Extreme Gradient Boosting (XGBoost) is a decision tree-based machine learning algorithm that builds predictive models using gradient boosting. It iteratively learns from the errors of weak learners (small decision trees) by training each tree to predict the residuals of the previous ones, thereby improving overall performance. XGBoost incorporates a regularized loss function to prevent overfitting, supports hyperparameter tuning, parallel processing, and handles various data types, including sparse data common in text vectorization. Widely used in text classification tasks, it has shown strong performance in studies such as Wicaksono et al. [31], who achieved up to 97.7% accuracy in judicial decision classification using TF-IDF, and Hendrawan et al. [32], where XGBoost outperformed Naive Bayes with an F1-score of 0.941 using Word2Vec.

Sentiment analysis is another area where XGBoost excels, as demonstrated by Samih et al. [33], who used XGBoost for sentiment classification through their Improved Word Vector for Sentiment Analysis (IWVS) model. This model generates enhanced word vectors by combining a polarized lexicon with the Sentiment2Vec technique. XGBoost is also utilized in other applications, such as Haumahu et al.'s [34] study on classifying Indonesian fake news, where XGBoost achieved 92% accuracy, showcasing its potential for distinguishing between true and false news.

XGBoost has proven effective in multi-label classification tasks, as demonstrated by Piter et al. [35], who used it to categorize scientific conference activities across diverse domains such as Economics, Science and Engineering, and Social Studies. The model successfully predicted a significant portion of the positive samples for each category, showcasing its robustness. In the criminal domain, Qi [36] applied XGBoost to classify theft-related crime reports using TF-IDF for feature extraction. A manually labeled dataset of 2,621 cases across seven theft categories was used, and XGBoost outperformed models like KNN, Naive Bayes, SVM, and GBDT, achieving precision of 96.8%, recall of 96.4%, and an F1-score of 96.6%.

Brandenburg [37] conducted a study on classifying Dutch police reports into four categories of online crimes (online threats, online distribution of sexually obscene imagery, computer trespassing, and general online crime) using machine learning techniques. Each class was treated as binary, allowing multi-label classification. After preprocessing and vectorizing the texts with TF-IDF, XGBoost was evaluated against Naive Bayes, Random Forest, and SVM. The results showed that XGBoost performed similarly without hyperparameter tuning and surpassed the other models when optimized, achieving F1-scores of 0.600 (General Online Crime), 0.540 (Online Threat), and 0.723 for both Obscene Imagery Distribution and Computer Trespassing, confirming its suitability for police report classification. The study demonstrated that XGBoost outperformed Naive Bayes, Random Forest, and SVM models, making it the most suitable for classifying police reports.

III. CRIME REPORT HIERARCHICAL MULTI-CLASS CLASSIFICATION

The method proposed in this work begins with the transformation of the raw database and proceeds through a series of preprocessing and modeling steps until the training phase. This entire workflow is illustrated in Figure 1. The process begins with a database consisting of crime incident report narratives. This database must include two main pieces of information: the unstructured text data representing the crime report narratives and the crime type classification provided by a human agent when the first information was initially checked. The method assumes that the crime type contains considerable distinct classes (i.e., a multi-class problem), making a naive Machine Learning approach challenging.

The first step in processing the database is to perform a data analysis. During this analysis, records with null fields should be discarded, as well as those that contain an insufficient number of characters. This minimum character count is necessary because, often, the police officer responsible for the report may only insert a period or a few characters to fulfill the field requirement. Once the dataset is cleaned, a study is conducted on the number of records in each available classification class. To do this, a table is constructed showing the count of records per class, along with the cumulative percentage, in order to identify the number of classes that represent the majority of the dataset. Based on a Pareto analysis of this table, only the most significant classes are selected, aiming to simplify the problem.

A stratified sample of the database, which preserves the distributions of the different classes, should be created and separated. This sample will be used at the end of the workflow to evaluate the models, ensuring that the data used in the evaluation is entirely unseen. With the remaining records and classes, a criminal area expert is consulted to develop a hierarchy of the possible classes. The construction of this hierarchy is based on the assumption that there are more related classes that can be grouped together. These groupings can be repeated at different levels, forming a classification tree.

The classification tree aims to replace a single model decision, which would involve a large number of classes, with a sequence of simpler decisions, as illustrated in Figure 2. These decisions are made by models specifically trained on subsets of the main dataset, which makes training faster and facilitates the interpretation of confusion matrices and ROC curves. The classification process follows a path through the tree's nodes, making successive decisions until it reaches one of the leaves, where the final classification is determined.

Once the tree is built, it is necessary to count the number of records at each decision node. The process starts at the leaves, where the records are counted. As we move up through the tree's levels, the number of records from the child nodes is added, forming the total count for each node. When we reach the root, it will contain the total number of records for the entire dataset.

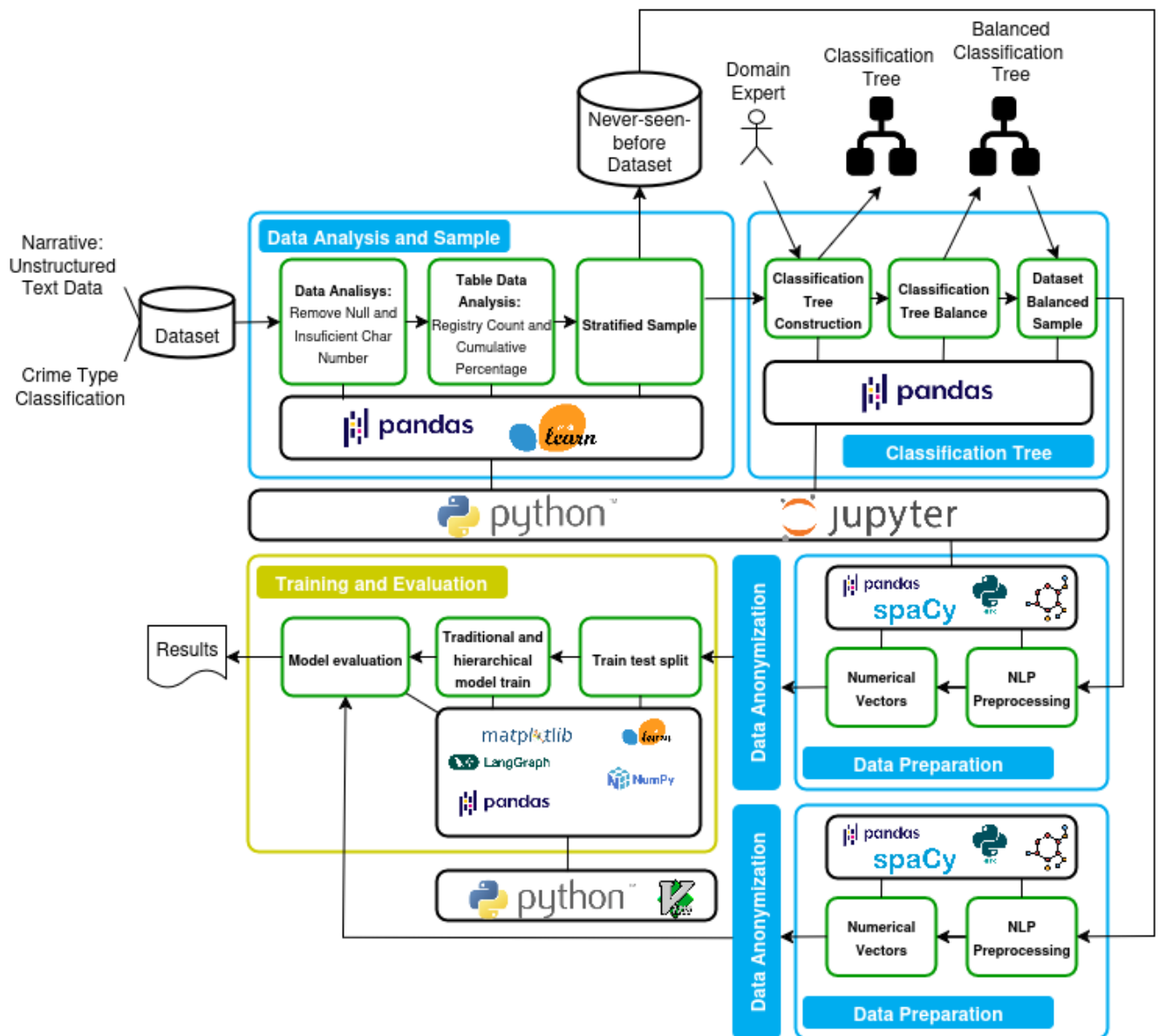


FIGURE 1. Architecture diagram of the case study. Green boxes represent methodology workflow steps, black ones represent technologies, blue ones represent CE1 (notebook), and the yellow one represents CE2 (Inmetro).

A bottom-up analysis is performed on the tree, traversing all possible paths. This analysis assesses whether all alternatives have the same number of records for each decision made by the model. If not, an under-sampling technique removes records from the majority class. After each under-sample, the counters are updated, propagating the changes to the root. At the end of the balancing process, all possible decisions will have the same number of records in their alternatives. It is important to note that, within the same tree level, one class may have more records than another as long as they do not participate in the same decision.

With the balanced classification tree, the leaf class counts are used to sample the original dataset. This procedure will result in a balanced dataset from the classification

tree. Next, the column referring to the crime incident report narratives undergoes Natural Language Processing (NLP) pre-processing to optimize the generation of semantic vectors. The pre-processing includes the following steps: conversion to lowercase, URLs removal, non-alphanumeric characters exclusion, digits and values elimination, tokenization, stopwords removal, lemmatization, and tokens re-grouping for use in embedding models.

Machine learning algorithms operate in numerical spaces, so textual data must be transformed into numerical vectors before processing. This transformation enables models to interpret and learn from language effectively. Unlike token-based methods, embedding techniques capture semantic relationships and contextual nuances, which enhances

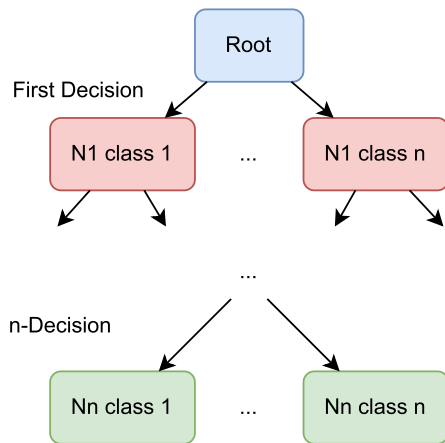


FIGURE 2. Classification tree for implementing hierarchical crime reports multi-class classification with n levels.

classification performance. Algorithms such as Word2Vec, TF-IDF, and word embeddings, among others, can be used in this process. The information regarding the crime incident report narratives is then replaced with the generated numerical vectors. Discarding the narrative column is crucial to ensure the dataset information privacy, as it is extremely difficult to reverse the process and recover the original data from the numerical vectors.

At this point, the dataset is ready to start model training. Before training, the dataset is stratified split into two sets: training and testing, following a commonly used proportion in the literature, such as 80% for training and 20% for testing. The test set will be reserved for evaluating each of the trained models. The training set will be used to train the models needed for decision-making. Model training involves searching for the best hyperparameters to achieve the best results according to a pre-established metric. During training, 5-fold cross-validation was performed using GridSearchCV to eliminate potential biases in the data.

A baseline model, referred to as the naive model, is trained to classify all classes in a single step, without incorporating any hierarchical structure. In contrast, the hierarchical model is built by training a separate classifier for each decision point in the classification tree. For each of these classifiers, a subset of the original dataset is selected, containing only the samples relevant to that specific decision. Each model then undergoes a hyperparameter optimization process to identify the most suitable configuration based on the chosen evaluation metric.

The hierarchical model is composed of a set of simple models. To perform a classification with this model type, the process begins by predicting the first class at the root of the classification tree and recording the prediction probability. Next, the path chosen is followed, and the model trained for the next decision is selected. The selected model then makes another prediction, recording its probability. This process is repeated until one of the leaves is reached. The final classification of the model corresponds to the leaf reached,

and the associated probability is calculated by multiplying the probabilities along the path taken.

An evaluation using a chosen metric is carried out after constructing both the naive and hierarchical models. The standard procedure for Machine Learning models is followed to evaluate the naive model, comparing the predictions with the actual values. For the hierarchical model, the evaluation is performed in a chained manner, considering each of the predictions along the path. The evaluation is based on the final classification obtained at the leaf reached for each input, comparing the predicted classification with the actual one. The stratified sample obtained at the beginning of the workflow must go through the calculation of semantic vectors and preprocessing and then be subjected to the tests for model evaluation.

IV. CASE STUDY EXPERIMENTS

The analysis of crime narratives, particularly in the context of government, plays a crucial role in understanding and preventing criminal behavior. Using data from a public safety agency, we propose a case study that explores how hierarchical text classification can significantly contribute to the organization, analysis, and interpretation of large volumes of information found in police reports. In a scenario where unstructured textual data accumulates on a massive scale, applying text classification becomes essential for extracting valuable patterns and insights. By focusing on a public safety agency that deals with a variety of daily and socially relevant incidents, we can evaluate the applicability and effectiveness of these techniques in a real and dynamic context.

We justify the choice of using data from a public safety agency to apply the hierarchical classification methodology by the opportunity to test and validate the solution in a complex environment with multiple layers of information. Crime narratives often involve nuances that can be captured by models capable of recognizing patterns of criminal events, as well as understanding the relationships between them. The case study thus provides a bridge between academic theory and practical application, enabling the findings of our work to have a direct and tangible impact on improving the operational processes of public safety agencies.

The Rio de Janeiro State Civil Police (SEPOL/RJ) is a government law enforcement agency in Rio de Janeiro State, Brazil. The agency operates several computational systems for recording crime incident report information, intelligence data, forensic analysis, and citizen-submitted complaints. The crime incident report dataset contains over 6.5 million entries from 2017 to 2024. It is important to note that the most crucial part of this dataset consists of unstructured data, including testimonies, statements, incident report narratives, intelligence reports, and complaint records, among others.

Text classification is of utmost importance for properly handling complaints made in the subsystem above and for the automatic preliminary categorization of offenses reported in incident reports. SEPOL/RJ handles around three thousand new crime incident reports statewide daily, spread across its

district and specialized police stations. These occurrences can be filed in person at the respective police stations or through the Online Occurrence Registration System (*Registro de Ocorrência Online* - RO Online). In the first case, occurrences must be manually classified and categorized by a police officer during the registration process. In the RO Online system, after the citizen registers the occurrence over the Internet, it is still analyzed like in-person filings, with no automation. The institution needs to process, group, and effectively utilize the unstructured data in its systems' diverse incident narratives.

Our case study began with a detailed analysis of the extensive crime incident report narratives dataset from SEPOL/RJ. The data was provided by SEPOL/RJ's Innovation Hub, ensuring the information's authenticity and accuracy. The dataset consists of 6,533,919 crime narratives, covering a variety of incidents recorded between 2017 and mid-2024. Each record was created or reviewed by a police officer, who was also responsible for classifying the occurrence into one of the 1,169 available categories of crime types.

A. ENVIRONMENT AND TOOLS

The computational infrastructure used for the experiments described in Section IV consisted of a laptop with a GPU and a server with two additional GPUs. For the entire workflow up to the generation of semantic vectors (embeddings), we employed an Acer notebook, model Predator Helios Neo 16, equipped with an Intel(R) Core(TM) i7-13650HX 13th generation processor, with 14 cores and 20 threads, 16 GB of RAM, and an NVIDIA GeForce RTX 4060 GPU with 8 GB of VRAM. From this point on, this computational environment will be referred to as CE1. On average, generating the embeddings took one hour for every 250,000 narrative conversions.

After generating the embeddings and removing the original crime incident report narratives, the datasets were transferred to a server provided by the Brazilian National Institute of Metrology, Quality, and Technology (Inmetro). The infrastructure available at Inmetro consisted of a DELL server equipped with two Intel(R) Xeon(R) Silver 4214 processors, each with 12 cores and 24 threads, running at 2.20 GHz, 256 GB of RAM, and two NVIDIA RTX 6000 Ada generation GPUs, each with 48 GB of VRAM. From then on, the computational environment will be referred to as CE2. This server was used for hyperparameter search, model training, and evaluation of the models' results.

We developed all the code for each step of the workflow using Python, as shown in Figure 1. On CE1, we employed the Jupyter programming environment using notebooks. On CE2, we adopted the native interpreter of the Linux operating system with the VIM text editor. The code was developed using the Pandas library, along with Numpy, Matplotlib, Scikit-Learn, and XGBoost. XGBoost was chosen for its robustness performance in sparse, high-dimensional data typical of text embeddings [36]. We employed this code for data analysis, dataset handling, and model training.

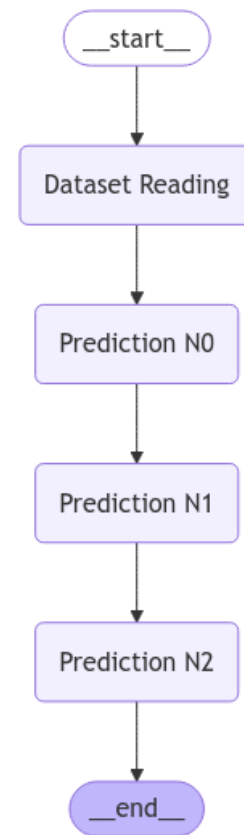


FIGURE 3. Graph for evaluating the hierarchical model.

For NLP preprocessing, we used the SpaCy library with the “pt_core_news-lg” corpus and Nltk. The Sentence Transformers library was used to load and utilize the “mE5-large-instruct” model for generating embeddings. The mE5-large-instruct model was selected for its superior multilingual performance in benchmarks, outperforming monolingual models in Portuguese [30].

We also use the LangGraph library to build a graph to perform the chained predictions of the hierarchical model, as shown in Figure 3. The constructed graph has four nodes: Dataset Reading, Prediction N0, Prediction N1, and Prediction N2. The first node reads the “never-seen-before” separated dataset for model evaluation and concatenates its fragments. The second node predicts the root of the class tree, determining the possible class for N1. Based on the prediction for N1, the next node selects the appropriate model to predict level N2. With the classifications of N1 and N2, the final node selects the most suitable model to predict the final class at level N3.

All the code developed during the experiments is publicly available in a repository on the GitHub service.¹

B. WORKFLOW APPLICATION

The workflow described in Section III was applied to the dataset. Starting with the data analysis, we removed reports

¹https://github.com/arvieira/HMTC_Crime_Report_ML

with empty text or filled with a single period character from the dataset, resulting in 6,174,717 records. In the table data analysis step, we counted the total records in each available crime type class. We applied a registry count approach to the records, computing each class's total number of occurrences. Then, we generated a table with each class's count and cumulative percentage, allowing us to identify the most and least frequent categories.

Based on this analysis, we observed that the 80 most frequent crime classes accounted for approximately 90% of all records. This insight guided our decision to filter the dataset and restrict the analysis to these 80 classes. The rationale behind this choice was to ensure that the classification models focused on the most representative and recurrent crime types, thereby improving model performance and interpretability while maintaining broad coverage of the dataset. This filtering step reduced the number of classes from 1,170 to 80, resulting in a minimal data loss of only 10% and leaving 5,536,041 records available for model training.

In this study, we adopted a snapshot approach, treating the entire dataset as if all records had occurred simultaneously. This timeless treatment was intentional, as the classification task focuses on the semantic structure of the crime narratives rather than temporal trends. During the data export process, certain columns were intentionally removed from dataset by the institution to prevent the use of unnecessary information, in accordance with data protection regulations. Consequently, year-to-year variations were not considered for model training or evaluation in this context.

We extracted a stratified sample from the original dataset, preserving each class's occurrence distribution. This sample represented 5% of the total records, totaling 276,803 entries. The sampling aimed to create a unique and "never-before-seen" dataset to evaluate the models developed during the study. After extracting the test sample, the dataset used for model construction remained with 5,259,238 records.

Considering that the classes in the dataset have a semantic relationship within the police/legal domain, we built a classification tree with three levels based on domain-specific knowledge. The structure was developed in collaboration with domain experts from the Civil Police of the State of Rio de Janeiro, ensuring that the grouping of classes reflected legal classifications and operational procedures. The hierarchy represents a consensus among specialists, aiming to preserve semantic coherence and practical applicability. We organized the 80 initial classes as leaves (N3) of the tree and created two additional levels (N2 and N1), with a root inserted at the top of the structure. This hierarchical organization is illustrated in Figure 2. Thus, to perform classification, the algorithm first decides at level N1, then at N2, and finally at N3, determining the final class for each crime incident report narrative. We represented the hierarchical structure of the classification tree by adding two new columns, N1 and N2, to the original dataset. The column that initially contained the crime type was renamed N3, which now represents the final level of the tree (the leaves).

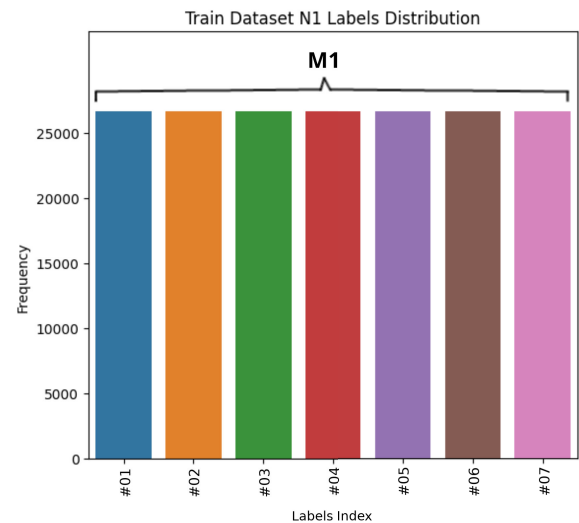


FIGURE 4. The chart shows the number of cases per class in the training dataset for level 1 of the classification tree. The name of the model (M1) trained to decide between the classes is highlighted. Labels index correspond to the tags listed in Figure 10.

We reviewed each decision that needed to be made in the classification tree to check for possible imbalances between the available classes. During this analysis, we identified an imbalance in the decisions that the models would make. In some cases, up to 97% of the records were concentrated in the majority class, while only 3% belonged to the minority class. This imbalance could introduce bias into the trained models because if the model decided to classify all occurrences as belonging to the majority class, it would achieve 97% accuracy, resulting in misleadingly high accuracy.

The undersampling procedure was applied specifically to balance the number of records across decision alternatives within the hierarchical classification tree. This technique ensures that both frequent and rare classes are equally represented during model training. By equalizing the number of samples per class at each decision point, the model is prevented from favoring majority classes, thereby preserving the importance and influence of rare categories in the learning process.

We structured the N1, N2, and N3 columns hierarchically to balance the dataset, with the tree leaves filled with the count of records for each N3 class. In the end, we balanced all possible decisions. Although nodes at the same tree level may have different counts, each decision ensures equal numbers of records across all possible alternatives. After balancing, the final set of records consisted of 186,549 records, which were randomly selected without replacement from the original dataset, maintaining the count in each leaf of the tree.

Following the dataset balancing strategy, Figures 4 and 5 illustrate the number of records per class used to train the models at levels 1 and 2 of the classification tree, respectively. These distributions reflect the balancing process, where each model was trained on a dataset containing an equal number

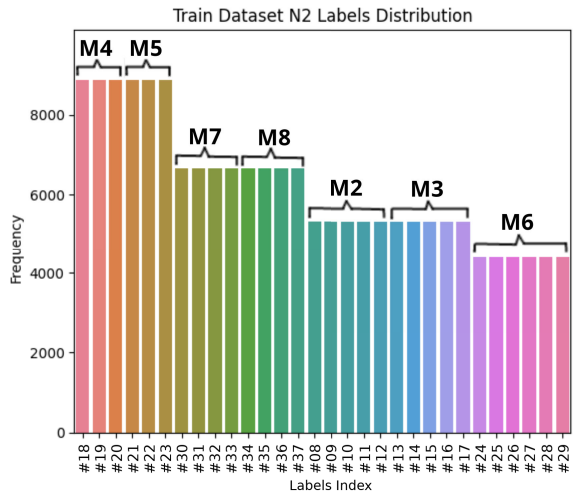


FIGURE 5. The chart shows the number of cases per class in the training dataset for level 2 of the classification tree. The name of the model (M1, M2, ...) trained to decide between the classes is highlighted. Labels index correspond to the tags listed in Figure 10.

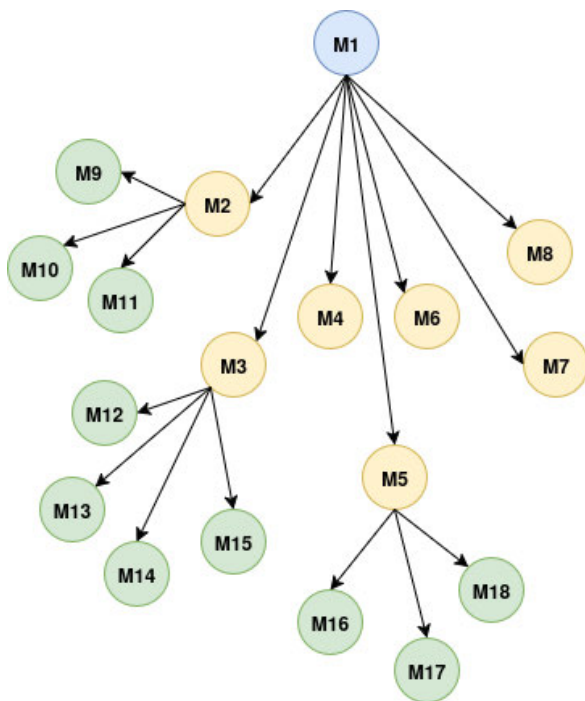


FIGURE 6. Display of the XGBoost model tree trained on the dataset, indexed according to the table in Figure 10.

of examples for each of the classes it was responsible for deciding. As a result, the models were not biased toward more frequent classes, ensuring fairer and more robust decision boundaries at each node.

Figure 10 complements this visualization by presenting a table listing all 18 XGBoost models trained, one for each decision node in the tree. It also shows the specific set of child classes that each model was designed to differentiate. Additionally, Figure 6 presents a graph illustrating the flow of decisions across the hierarchical model. For instance,

depending on the output of model M1 at level N1, the next classification step is performed by one of the models M2 through M8. If M1 predicts Property Crimes (*Crimes Contra a Propriedade*), the next iteration will be handled by model M2. Subsequently, based on M2's decision, the final classification is performed by one of the models M9, M10, or M11. The mapping between classes, model names, and their respective child nodes is detailed in Figure 10. Together, these figures demonstrate how the classification problem was decomposed into a series of balanced and structured multiclass decisions.

The narratives of the incident reports are composed of free text written or reviewed by police officers during their service to the population of Rio de Janeiro State. These texts underwent a preprocessing process to optimize the results of the semantic vector generation model. The preprocessing involved the following steps: lowercase conversion, URLs removal, non-alphanumeric characters exclusion, digits and values elimination, tokenization, stopwords removal, lemmatization, and token re-grouping for use in embedding models. At the same time, we created a copy of the dataset without the preprocessing transformations to serve as a comparison base for the results obtained.

After generating the preprocessed and raw datasets, we employed the “mE5-large-instruct” model to generate their respective semantic vectors (embeddings), consisting of 1024 values. We discarded the narrative column from the datasets to preserve potential personal data, keeping only the 1024 columns of the semantic vector and the three columns corresponding to the classification tree. The datasets were divided into training and testing sets in an 80-20 ratio. We used a label encoder to encode the classes to numeric values in the datasets. From the training set, we performed grid searches with stratified cross-validation to optimize the hyperparameters of the XGBoost model for all decisions at the N1, N2, and N3 levels. The best hyperparameters were then applied to train models for each of the eighteen possible decisions in the class tree. Thus, we obtained an optimized hierarchical ensemble model to predict each node at levels N1, N2, and N3.

Predicting a record's class involves using the root model to select a node at N1. After this choice, depending on the node selected at N1, a distinct model specific to the decision would be employed to determine the class at N2. This process was repeated for the selection of the class at N3. The procedure would only be successful if the choices at all three levels (N1, N2, and N3) were made correctly. Thus, the accuracy of the hierarchical model is calculated as the ratio of correct predictions to the total number of cases at N3. The model's confidence is determined by multiplying the probabilities of the decisions made at the three levels (N1, N2, and N3).

C. NAIVE MACHINE LEARNING APPROACH AS A BASELINE

The naive ML approach for classification involves training a model on a dataset that contains all possible classes in the

label column. During training, the model learns to distinguish between these classes to later classify new data. Although this approach is widely used, it presents difficulties when the number of classes is very large. With too many classes, the model struggles to generalize correctly, leading to a significant increase in errors, especially when some classes are similar or have few examples.

Additionally, the large number of classes can result in a more time-consuming and complex training process. The model has to learn to differentiate between a wide range of categories, which can overload the learning process and reduce accuracy. This becomes a significant challenge, especially in scenarios where the available data is limited or imbalanced across the classes. The model might end up “confusing” classes with similar characteristics, negatively impacting its performance.

In this context, we use the naive approach as a baseline for comparison with the approach proposed in this paper. The idea is to use the 80-classes classification as a starting point, so we can observe how class hierarchy improves the model’s performance. With the hierarchy, the model trains on smaller subsets, with fewer classes at a time, which increases the chance of correct classification and simplifies the learning process. This way, we can compare the model’s accuracy and F1-score with hierarchically organized classes and check if the proposed approach yields better results than the naive model.

D. PARAMETERS AND METRICS

The experiment’s main objective was to analyze and compare the performance of a hierarchical model against a naive flat ML model to understand the benefits and limitations of each approach. A naive flat ML model consists of the XGBoost algorithm trained with the entire dataset to predict the 80 possible classes. For this evaluation, we applied a systematic model optimization approach, focusing on selecting the most effective parameters for both the hierarchical and naive models. The parameter search process involved multiple rounds of grid search, using 5-fold cross-validation. In each search round, the most promising parameters identified were carefully analyzed and used to adjust the next search, allowing the parameter grid to be gradually and efficiently adapted. This continuous refinement ensured that the search for better parameters became increasingly accurate until it was possible to identify a configuration that maximized the model’s accuracy. Once the parameter grid could no longer be significantly adjusted, the resulting matrix was considered the ideal configuration. All the hyperparameter search grids used in the training of each model are publicly available in a repository on GitHub.²

We optimized the process to train the XGBoost model, which aimed to predict one of the 80 classes in the dataset and to train each node model in the classification tree. We adapted the grid search to meet the specific needs of

each node prediction, considering the particularities of the hierarchical model and the structure of the classification tree. For the evaluation, we used the confusion matrix, which allowed a detailed analysis of the model’s performance by correctly identifying the predicted and misclassified classes. As summary metrics, we chose accuracy and F1-score, both commonly used in classification problems to assess overall correctness and balance between precision and recall. Additionally, we applied the Wilcoxon signed-rank test to compare the F1-score performance per class between models, allowing us to assess the statistical significance of the observed improvements. This approach provides additional insights into the model’s quality and areas for potential improvement. In addition to predicting the final class, each model, both the naive and the hierarchical approaches, provided a vector containing the probabilities associated with each possible class.

E. RESULTS AND DISCUSSION

The experiments conducted on CE2’s server infrastructure aimed to compare the performance of two models based on the XGBoost algorithm. The baseline model consists of a single model trained to predict one of the dataset’s 80 possible classes. The hierarchical model implements our proposal in a combination of multiple models (ensemble), responsible for deciding each level of the classification tree described in Section III.

It is worth noting that the number of possible options at each node of the classification tree is much smaller than the 80 available classes, as it follows a hierarchical structure. The model trained for that specific level at each decision node only chooses among its possible alternatives. Each model returns the predicted class and the prediction probability associated with that class. Finally, when the graph reaches the last node, the probabilities of each decision made along the way are multiplied, resulting in the hierarchical model’s final prediction probability.

Only the database composed of semantic vectors generated from the raw text of the criminal report dynamics was used for the tests. Applying the preprocessing described in Section III led to a lower performance in the accuracy metric. It is important to emphasize that the need to preprocess the dataset before generating the embeddings is directly related to the type of model that will be used for this task. In the specific case of the “mE5-large-instruct” model, preprocessing the dynamics resulted in lower-quality semantic vectors for training algorithms like XGBoost, compromising the final accuracy.

After conducting the CE2 experiments, we observed that the naive model achieved an accuracy of 0.419 when evaluated across 80 distinct classes. This value served as a baseline for performance comparison with the proposed hierarchical model. On the other hand, the hierarchical model reached an accuracy of 0.463, surpassing the baseline by almost 5%, representing a relative gain of 12%. To complement this analysis, we also evaluated the models

²https://github.com/arvieira/HMTC_Crime_Report_ML

using the weighted F1-score, a more robust metric for imbalanced multi-class problems. The naive model obtained a weighted average F1-score of 0.44061, while the hierarchical model achieved 0.48188, representing an improvement of approximately 9.4%. A Wilcoxon signed-rank test was applied to the F1-score per class to assess the statistical significance of the observed improvements, yielding a p-value of 0.000010. The models' decisions must cover all 80 possible classes. Thus, considering the hypothetical use of a completely random model in a balanced dataset scenario, its accuracy would be $1/80 = 0.0125$. This value highlights the importance of improving the accuracy. Given the large number of classes, any increase would be a significant advancement.

Furthermore, a significant reduction was observed in the time required to complete the models' training. The naive model, which needs to distinguish between 80 different classes, took 1.4 minutes to complete each round of cross-validation. In contrast, although the hierarchical model consists of 18 distinct models, it still demonstrated a shorter total training time, with a reduction of 25.48% compared to the baseline model. It is important to emphasize that both models were trained using the same dataset containing the same number of records at the beginning of the process. This result highlights the hierarchical model's higher computational efficiency.

When analyzing the resulting confusion matrices, both for the naive model with 80 classes and for each step of the hierarchical model, it becomes clear that the hierarchical model offers greater clarity and practicality in interpreting the results. The confusion matrix of the naive model, which has 80 rows and 80 columns, becomes difficult to read and analyze efficiently, as shown on Figure 7. The same applies to the analysis of ROC curves. Analyzing such a curve can indicate how well a model can discriminate one class from the others. However, breaking the problem into hierarchical levels allows the evaluation of the model's performance separately at each decision point during the prediction process, analyzing the descent path through the classification tree.

As an example of analysis, we can evaluate the confusion matrix of one of the models trained for the hierarchical approach. The model chosen was the one trained to classify the leaves of the classification tree for the node at level N2 related to robberies. Figures 8 and 9 illustrate the confusion matrix and the ROC curve for this model, respectively.

This matrix consists of 10 rows and 10 columns, representing all the classes associated with robbery. By analyzing this matrix, at least two insights can be drawn.

The confusion matrix row referring to the class Robbery (Other) (*Roubo outros*) shows many predictions being incorrectly distributed among other classes. This may raise a flag for SEPOL/RJ, as it could indicate that many crime narratives were classified into a generic robbery class when they could have been classified into a more specific one. Due to police officers' fatigue and stress, as well as the emotional

Confusion Matrix Heatmap

80-Class Naive Model

0 16,000

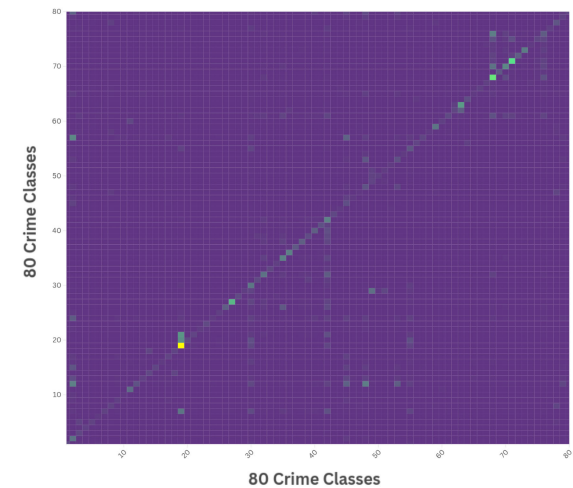


FIGURE 7. Confusion matrix heatmap of the 80-classes model.

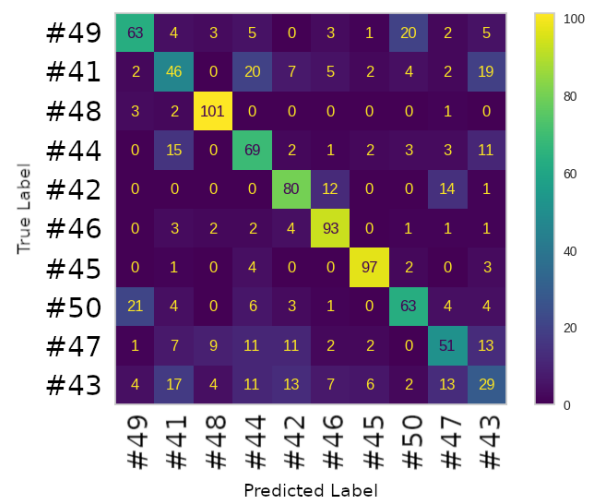


FIGURE 8. Confusion matrix of the robbery model. Labels index correspond to the tags listed in Figure 10.

state of crime victims requiring fast service, more generic classes may have been selected in an effort to provide a faster response. A more in-depth study of the behavior of those involved and the design of the SEPOL/RJ's system classes could be developed in the future based on this point raised by the confusion matrix analysis.

Another valuable insight in the analysis of Figure 8 is the row referring to Robbery Inside Vehicle (*Roubo no Interior de Veículos*), which shows a high number of errors, with many incorrect predictions for the classes Cell Phone Robbery (*Roubo de Telefone Celular*) and Vehicle Robbery (*Roubo de Veículo*). Considering the problem domain, further analysis shows that robberies inside vehicles often involve the theft of cell phones, which could confuse the classifications.

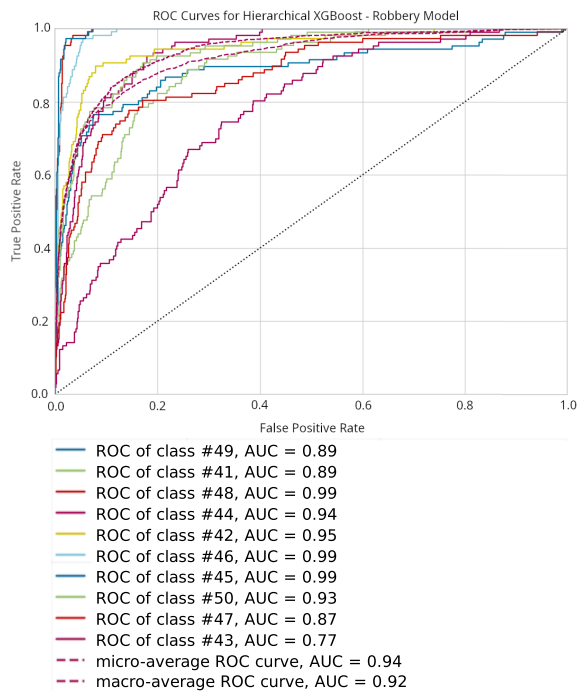


FIGURE 9. ROC Curves of the robbery model. Labels index correspond to the tags listed in Figure 10.

Additionally, the similarity between the keywords used in the crime narrative of Vehicle Robbery (*Roubo de Veículo*) and Robbery Inside Vehicle (*Roubo no Interior de Veículos*) may lead to overlap in the semantic vectors causing the model to struggle with correctly distinguishing them.

Other valuable insights can be obtained from the analysis of the confusion matrices and ROC curves generated during the training of the hierarchical approach models. All Figures are publicly available in a repository on the GitHub service.³

It is important to emphasize that splitting the problem into hierarchical levels risks losing crucial information about false positives that occurred at earlier levels. For example, by analyzing only the confusion matrix at level N3, as done in the Robbery (*Roubo*) analysis, the opportunity to identify cases where the initial classification at higher levels of the class tree was incorrect is lost, but the error only manifested in subsequent decisions. Therefore, conducting a complete analysis of the path through the tree is essential, starting with the confusion matrix at the root and following each decision to the lower levels.

Another relevant aspect of the hierarchical model is the generation of a prediction probability vector. With the prediction probability vector produced, a user can set a confidence level for the prediction made. If the model's confidence is below a predefined threshold, it can simply refrain from providing a classification, indicating that it is

uncertain about the result. This feature allows the model to be used more assertively, functioning as a “co-pilot” for police officers. In this format, the model would only suggest a type of crime when it had a high degree of confidence in the prediction. This would significantly reduce errors caused by incorrect predictions, providing greater trust in the decisions based on the AI's results.

When comparing the results of this work with other recent studies in the field of hierarchical text classification, it is important to consider both performance metrics and the nature of the datasets involved. Kusakin et al. achieved higher F1-score values than those obtained in this work by employing a domain-specific BERT model to classify scientific texts in Russian. However, their dataset consisted of structured, well-written academic content, which contrasts sharply with the unstructured, often emotionally charged and inconsistently written narratives found in criminal incident reports. Garcia et al. proposed a hierarchical classification framework for scholarly publications and reported a best-case weighted F1-score of 0.5973. Among the transformer-based models they evaluated, BERT achieved an F1-score of 0.5582, GPT-4 reached 0.3390, and LLaMA 3.1 obtained 0.3311. While these results also surpass our model's weighted F1-score of 0.48188, it is essential to highlight that, similarly to Kusakin et al., Garcia et al. worked with scientific texts that follow formal language patterns and domain-specific vocabulary. In contrast, our model was trained on real-world crime narratives from the Civil Police of the State of Rio de Janeiro, which are often informal, fragmented, and contextually complex, produced under stressful conditions by victims or law enforcement officers. Despite the challenging nature of our dataset, our hierarchical model showed statistically significant improvements over the baseline, as confirmed by a Wilcoxon signed-rank test (p -value = 0.000010), demonstrating the robustness of our approach in a highly noisy and operationally critical domain.

It is also worth highlighting the performance of transformer-based models evaluated by Garcia et al. ([8]) within their classification framework. Specifically, BERT achieved an F1-score of 0.5582, while GPT-4 and LLaMA 3.1 reached 0.3390 and 0.3311, respectively. Although BERT slightly outperformed our model in terms of F1-score, both GPT-4 and LLaMA 3.1 underperformed when compared to our hierarchical approach, which obtained a weighted F1-score of 0.48188. These results suggest that, in certain scenarios and domains, especially those involving complex or noisy textual data, such as criminal incident narratives, a well-structured hierarchical classification strategy using traditional models like XGBoost can outperform even state-of-the-art transformer-based architectures. This reinforces the importance of tailoring classification approaches not only to the algorithmic sophistication, but also to the specific characteristics and challenges of the dataset, the availability of specialized hardware for training large models, and the privacy-sensitive nature of the data involved.

³https://github.com/arvieira/HMTC_Crime_Report_ML

Model Name	Node Name	Tag	Node Children
M1	Root	#01	Property Crimes (M2)
		#02	Crimes Against Persons (M3)
		#03	Traffic or Environmental Crimes (M4)
		#04	Vehicle Recovery, Administrative Acts, or Incident Reporting (M5)
		#05	Drug, Narcotics, and Firearm Possession Related (M6)
		#06	Resistance, Contempt, or Disobedience (M7)
		#07	Trespassing, Disturbance, Damage, or Arbitrary Exercise of Rights (M8)
M2	Property Crimes	#08	Fraud (M9)
		#09	Robbery (M10)
		#10	Theft (M11)
		#11	Receiving Stolen Goods (Leaf)
		#12	Embezzlement (Other) (Leaf)
M3	Crimes Against Persons	#13	Homicide (M12)
		#14	Rape (M13)
		#15	Bodily Injury (M14)
		#16	Threat, Insult, Stalking, Damage, or Extortion (M15)
		#17	Violation of Urgent Protective Measures (Leaf)
M4	Traffic or Environmental Crimes	#18	Tampering with Vehicle Identification Marks (Leaf)
		#19	Driving Without License or Prohibited from Driving (Leaf)
		#20	Criminal and Administrative Sanctions on the Environment (Leaf)
M5	Vehicle Recovery, Administrative Acts, or Incident Reporting	#21	Vehicle Recovery (M16)
		#22	Administrative Acts (M17)
		#23	Incident Reporting (M18)
M6	Drug, Narcotics, and Firearm Possession Related	#24	Drug Trafficking (Law 11.343/06) (Leaf)
		#25	Possession of Drugs for Personal Use (Law 11.343/06) (Leaf)
		#26	Drug Trafficking Association (Law 11.343/06) (Leaf)
		#27	Seizure of Narcotic Substance (Leaf)
		#28	Drug Law (Other) (Law 11.343/06) (Leaf)
		#29	Illegal Possession of Restricted Firearm (Leaf)
M7	Resistance, Contempt, or Disobedience	#30	Contempt (Leaf)
		#31	Resistance (Leaf)
		#32	Disobedience (Leaf)
		#33	Aggravated Resistance (Leaf)
M8	Trespassing, Disturbance, Damage, or Arbitrary Exercise of Rights	#34	Trespassing (Leaf)
		#35	Disturbance of Peace (Leaf)
		#36	Arbitrary Exercise of One's Own Rights (Leaf)
		#37	Damage (Other) (Leaf)
M9	Fraud	#38	Fraud (Other) (Leaf)
		#39	Attempted Fraud (Other) (Leaf)
		#40	Credit Card Fraud (Leaf)
M10	Robbery	#41	Robbery of Pedestrian (Leaf)
		#42	Vehicle Robbery (Leaf)
		#43	Robbery (Other) (Leaf)
		#44	Cell Phone Robbery (Leaf)
		#45	Robbery Inside Public Transport (Leaf)
		#46	Motorcycle Robbery (Leaf)
		#47	Robbery Inside Vehicle (Leaf)
		#48	Cargo Robbery (Leaf)
		#49	Robbery of Commercial Establishment (Leaf)
		#50	Robbery Inside Commercial Establishment (Leaf)
M11	Theft	#51	Theft (Other) (Leaf)
		#52	Cell Phone Theft (Leaf)
		#53	Theft from Pedestrian (Leaf)
		#54	Vehicle Theft (Leaf)
		#55	Theft Inside Residence (Leaf)
		#56	Theft Inside Commercial Establishment (Leaf)
		#57	Theft Inside Public Transport (Leaf)
		#58	Theft from Commercial Establishment (Leaf)
		#59	Theft Inside Vehicle (Leaf)
		#60	Motorcycle Theft (Leaf)
		#61	Bicycle Theft (Leaf)
		#62	Vehicle Plate Theft (Leaf)
M12	Homicide	#63	Homicide by Firearm Projectile (Leaf)
		#64	Attempted Homicide by Firearm Projectile (Leaf)
M13	Rape	#65	Rape (Leaf)
		#66	Rape of a Vulnerable Person (Leaf)
M14	Bodily Injury	#67	Bodily Injury (Other) (Leaf)
		#68	Bodily Injury Caused by Punches, Slaps, and Kicks (Leaf)
		#69	Unintentional Bodily Injury (Other) (Law 9503/97) (Leaf)
		#70	Unintentional Bodily Injury Caused by Vehicle Collision (Leaf)
		#71	Unintentional Bodily Injury Caused by Run-Over (Leaf)
		#72	Bodily Injury Caused by Firearm Projectile (Leaf)
		#73	Physical Altercation (Leaf)
M15	Threat, Insult, Stalking, Damage, or Extortion	#74	Threat (Leaf)
		#75	Insult (Other) (Leaf)
		#76	Insult Due to Prejudice (Leaf)
		#77	Defamation (Leaf)
		#78	Slander (Leaf)
		#79	Mistreatment (Leaf)
		#80	Stalking Crime (Art. 147-A of Penal Code) (Leaf)
		#81	Extortion (Other) (Leaf)
M16	Vehicle Recovery	#82	Recovery of Stolen Vehicle (Leaf)
		#83	Recovery of Stolen Vehicle (Theft) (Leaf)
M17	Administrative Acts	#84	Precautionary Measure for Future Rights (Leaf)
		#85	Execution of Arrest Warrant (Leaf)
		#86	Seizure (Other) (Leaf)
		#87	Vehicle Seizure (Leaf)
		#88	Removal for Death Verification (Leaf)
		#89	Seizure of Juvenile Offender (Articles 104 and 107 of Law 8.069/90) (Leaf)
		#90	Resulting from Preventive Detention by Police Unit (Leaf)
M18	Incident Reporting	#91	Atypical Fact (Leaf)
		#92	Document Loss (Leaf)
		#93	Cell Phone Loss (Leaf)
		#94	Death Without Medical Assistance (Leaf)
		#95	Vehicle Plate Loss (Leaf)
		#96	Found Missing Person (Leaf)
		#97	Disappearance (Other) (Leaf)

FIGURE 10. Table of XGBoost models trained at each node of the tree, along with the child classes among which they had to decide. Class labels are color-coded: blue for level N1 decisions, yellow for level N2, and green for level N3 (leaf nodes).

V. CONCLUSION AND FUTURE WORKS

This work proposed a new workflow to the multi-class text classification problem, applied to the crime incident report narratives scenario. The approach is especially recommended for problems with a large number of possible classes, where naive methods face difficulties during training and model convergence. The solution involves constructing a classification tree based on domain knowledge, which organizes all possible classifications in a hierarchical manner. Using classification trees, models are trained for each possible decision. Thus, to make a prediction, the model follows the path from the root to one of the leaves of the classification tree.

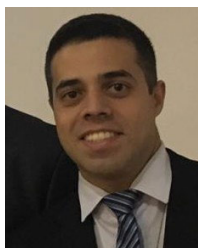
The results showed a 5% improvement in accuracy for the hierarchical model, which achieved 0.463 compared to the baseline model's 0.419, along with a 25.48% reduction in training time. By hierarchizing the classes, the problem was divided into multiple subproblems, replacing a single model for 80 classes with 18 models, each handling fewer than 10 classes. This structure facilitated the analysis of simpler models through confusion matrices and ROC curves, offering deeper insights into the models' behavior and providing valuable information for public administration. Additionally, the hierarchical model achieved a weighted average F1-score of 0.48188, compared to 0.44061 for the baseline model, reinforcing its superior classification performance. A Wilcoxon signed-rank test applied to the F1-score per class produced a p-value of 0.000010, confirming the statistical significance of the improvements. Furthermore, the hierarchical model incorporated prediction probability, allowing public administration to set a confidence threshold for classification, ensuring that predictions are only made when the model's confidence is sufficient, thus minimizing the risk of errors in crime incident record databases.

Future work will focus on evaluating other ML models besides XGBoost and using alternative hyperparameter optimization techniques, such as Optuna. It would be interesting to investigate the behavior of different models when faced with partitioning problems with many possible classes. Using alternative metrics, such as the Matthew Correlation Coefficient and Cohen's Kappa instead of accuracy, could provide a deeper analysis of the hierarchical model's performance. Additionally, we plan to explore deep learning techniques—especially large language models (LLMs)—to enable hierarchical classification without depending on word embeddings. A promising advancement in this line of research is the integration of multi-agent systems, where LLM-based agents are equipped with tools composed of previously trained traditional hierarchical ML models. These tools would allow the agents to make accurate predictions in complex classification tasks with many possible classes, enhancing both the scalability and intelligence of the overall system.

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